A complete kernel-only Lean proof of the three-copy Werner endpoint for every dimension

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Abstract

For C an operator on $(\mathbb{C}^d)^{\otimes 3}$ of rank at most 2, let $q(C) = \sum_{S\subseteq\{1,2,3\}}(-1/2)^{|S|}\|\mathrm{Tr}_S\, C\|_F^2$. The inequality $q(C) \geq 0$ is the three-copy Werner endpoint. I prove nothing new about Werner states in this note: the endpoint was already proved for every $d \geq 3$ by Tomar [1], whose Lean formalization covers $3 \leq d \leq 8$ only. What I contribute is a complete kernel-only Lean 4 proof of the endpoint for every d, by an independent, self-contained route: one exact rational certificate of the $t = 1/d \to 0$ limit of a strengthened moment relaxation, carried to every dimension by a lemma chain, with the certificate's exact identity proved in the Lean kernel. The final theorem depends only on the standard axioms `propext`, `Classical.choice`, and `Quot.sound`, with no `sorry` and no `native_decide`, and I had its statement independently checked to confirm it says what I claim here.

1 Introduction

The Werner state $\rho_\alpha \propto I - \alpha F$ on $\mathbb{C}^d \otimes \mathbb{C}^d$, where F is the swap, is a central test case for distillability. For k copies, undistillability reduces to a quadratic-form inequality on operators C of rank at most 2 on $(\mathbb{C}^d)^{\otimes 3}$ when $k = 3$. Define, for $\alpha \in \mathbb{R}$,

$$q_\alpha(C) = \sum_{S\subseteq\{1,2,3\}} (-\alpha)^{|S|}\|\mathrm{Tr}_S\, C\|_F^2, \qquad q = q_{1/2}. \tag{1}$$

Then ρ_α is three-copy undistillable exactly when $q_\alpha \geq 0$ on rank-≤ 2 operators C, and the continuation identity used in the earlier work on this problem shows that $q_\alpha \geq (1-2\alpha)(1-\alpha)^2\|C\|^2$ on $[0, 1/2]$ holds in dimension d exactly when the endpoint $q \geq 0$ holds there. The endpoint is therefore the sharp three-copy statement.

Tomar [1] proves the endpoint for every $d \geq 3$, by a base certificate polynomial in $1/d$ for $d \geq 9$ plus exact certificates for smaller d. That paper is a preprint, not peer reviewed, and its Lean formalization covers $3 \leq d \leq 8$ only: the paper states that "the case $d \geq 9$ is not formalized". I close that formalization gap, by a different certificate route, and I claim nothing else as new. This proof does not use Tomar's certificate or argument as a step; it proves the endpoint from the $t = 0$ certificate and lemma chain in this note.

Theorem 1. *For every d and every operator C on $(\mathbb{C}^d)^{\otimes 3}$ of rank at most 2, $q(C) \geq 0$.*

Theorem 1 is Tomar's theorem, restated. My contribution is Theorem 2 below: the same statement, proved in Lean 4 by the kernel, for every d at once.

2 The moment relaxation and its $t = 0$ limit

I prove Theorem 1 in Lean through a strengthened moment relaxation. A rank-≤ 2 operator C lifts to moments built from its factors; twirling over the relevant symmetries lands in a commutative algebra spanned by diagram operators, and the question becomes whether a certain moment functional is nonnegative on the physical moment set. Replacing the orbit variables by trace coordinates m makes the relaxation d-free except for the site modules: the objective and the equality rows are trace functionals, so they have d-free coefficients in m, while only the positive-semidefinite blocks depend on d, through site matrices $N_r(p)(t)$ with $t = 1/d$.

At $t = 0$ no site module degenerates: every module's limit matrices span the full matrix algebra. An exact rational certificate $X_b \succeq 0$ and multipliers λ for the $t = 0$ relaxation therefore suffices, by the lemma chain of Section 3, to prove $q(C) \geq 0$ in every dimension. The certificate I use is the compact v2 certificate shipped in this folder. It passes the standalone exact verifier `verify_t0.py`: 1,000 blocks are positive definite. It is 11.1× smaller in bits than the earlier v1 certificate (2.11M bits versus 23.48M bits), and its correction denominator is 131 bits (v1: 5,941 bits). These are properties of the shipped certificate data, checked by exact rational arithmetic outside Lean; Section 4 is what moves the same identity into the Lean kernel.

3 The lemma chain

The chain is an abstract theorem in Lean, `endpoint_of_t0_cert`: given a $t = 0$ certificate, the endpoint follows in every dimension. Its steps are:

1. *Embedding.* A physical point of dimension e embeds into $d \geq \max(e, 4)$ without changing m or q, because the permutation operators commute with the embedding and partial traces commute with it.

2. *Validity at dimension d.* The twirled coordinates satisfy every homogeneous row, the trace row equals 1, the objective equals $q(C)$, and every word block and coupled block is positive semidefinite, for any real tensor-space module basis.

3. *d-freeness in m.* Every row and the objective are trace functionals, so their m-forms are exactly their d-pipeline forms composed with the Weingarten change of variables. The qubit factor causes no gap, since the conjugated moments lie in the range of the qubit Gram matrix.

4. *Blocks at $t = 1/d$.* For $d \geq 4$, the blocks built from $N_r(p)(1/d)$ are positive multiples of compressions of the true blocks: the site data are positive multiples of Gram-form matrices, the one-site Gram matrix at the identity is positive definite, and the denominator $(1-d^{-2})(1-4d^{-2})(1-9d^{-2})$ is positive. Hence the blocks are positive semidefinite for every physical m of dimension $e \leq d$.

5. *Continuity.* The site matrices are polynomial in t, so the blocks converge as $t = 1/d \to 0$, and the positive-semidefinite cone is closed. Every physical m, of every dimension, therefore lies in the $t = 0$ feasible set.

6. *Pairing.* At a physical m, $q(C) = \sum_b \langle X_b, L_b(0)(m)\rangle + \lambda_{\text{trace}}$, a sum of nonnegative pairings plus a nonnegative trace multiplier. Hence $q(C) \geq 0$.

I proved each step in Lean in the `AllD` layers, with no `sorry` and only the standard axioms. The chain is generic: it does not depend on the particular certificate, only on the existence of one satisfying the $t = 0$ certificate conditions. The remaining work is therefore exactly one thing: proving the certificate identity for the explicit v2 data, in the kernel.

4 Layer D: the certificate identity in the kernel

The one hypothesis of `endpoint_of_t0_cert` that is not generic is the exact identity for the explicit certificate data, `IdentityT0`: for every orbit-invariant μ, the objective equals the sum over blocks of $\langle X_b, L_b(\mu)\rangle$ plus the row terms. I prove this identity in the Lean kernel, in three parts.

First, the *bridge* rewrites each block pairing as a sum over family parts, with a common integer scale, so that the identity becomes a vanishing statement for family totals. Second, the *glue* reduces that statement to `FamZero` for the three families, i.e. to literal-free family totals, using only the orbit invariances of μ; coupled block parts fold through a kernel-checked full-rank re-presentation. Third, the *compute* checks the family totals in the kernel, in 8,199 generated identity modules pinned by SHA-256 in `identity_manifest.json`, together with the hashes of the four certificate files (`blocks.jsonl`, `multipliers.json`, `qubit.json`, `sites.json`). A rep-budget split was needed to fit the largest claim modules under the process memory limit; it raised the module count from 8,110 to 8,199 without changing any family total.

The complete build ran as run 41356979845, in one 8-CPU, 9 GB container, from a clean checkout, starting 2026-11-01 22:26:47 UTC and finishing 2026-11-02 10:29:39 UTC (12 h 3 min wall time, 18.0 CPU-hours summed over the modules). All 8,199 modules kernel-checked with return code 0 (prep 9, light 7,829, heavy 145, final 216); 62 attempts of the parallel light pass were killed for memory and passed in the serial retry; the run summary records no Lean error anywhere. Peak memory was 6.40 GB for one module.

Theorem 2. *In Lean 4, with Mathlib, the theorem* `LayerDT0.werner3_endpoint_all_d` *states that for every* $d : \mathbb{N}$ *and every* $C : \mathrm{Matrix}\,(\mathrm{Fin}\,3 \to \mathrm{Fin}\,d)\,(\mathrm{Fin}\,3 \to \mathrm{Fin}\,d)\,\mathbb{C}$ *with* $C.\mathrm{rank} \le 2$, $0 \le \mathrm{qfun}\,C$, *where* `qfun` *is the Lean definition of* q. *It depends only on the axioms* `propext`, `Classical.choice`, *and* `Quot.sound`.

The same axiom list holds for `LayerDT0.identityT0`. I scanned every shipped Lean file for `sorry`, `admit`, `axiom`, `native_decide`, `implemented_by`, `extern`, `unsafe`, and `opaque` outside comments; the scan found none (see `tests/test_lean_scan.py` in this folder). An independent agent that had not seen the work producing the proof checked the final theorem's statement and rated it faithful: `qfun` is proved equal in Lean to an independently written q (the six proper partial traces written out on index triples, Mathlib's `Matrix.trace` for the full trace, `Complex.normSq` for squared entries), rank is Mathlib's `Matrix.rank` with no coercions, no shadowing declarations or extra axioms were found in the import chain, $d = 1$ gives $q = \|c\|^2/8$ in Lean as expected, and an independent NumPy comparison on random complex C gave maximum relative difference approximately $3e-15$. The proof term itself was out of scope for that check; the kernel build above is what checks it.

5 Prior art, and what I do not claim

Tomar [1] proves Theorem 1 for every $d \ge 3$. What is new here is Theorem 2: as far as I have found, the first complete kernel-only Lean proof of the endpoint for every d, closing the formalization gap at $d \ge 9$ left by Tomar [1]. If an earlier complete kernel-only Lean proof for every d exists, I ask to

be told, and I will correct this claim. Nothing here concerns any other number of copies, and my claim is the all-d proof only.

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I did this work with substantial help from AI models (Anthropic's Claude): searching, writing code and Lean proofs, and running the independent reviews described above. I directed the work and I am responsible for every claim; that is why everything here is backed by code, data, tests, and machine-checked proofs that anyone can rerun from this folder alone.

References

[1] Tejasvi Singh Tomar. Undistillability of werner states from three, four and five copies. Zenodo preprint, version 4, October 2026, not peer reviewed. URL https://doi.org/10.5281/zenodo.23718035. ORCID 0000-0003-4668-3639. Proves the three-copy endpoint for every $d \geq 3$; Lean formalization covers $3 \leq d \leq 8$ only, with the case $d \geq 9$ stated as not formalized.